

PROFIL

Embedded Software Engineer with experience in automotive Test & Validation, specializing in C/C++, embedded systems, and hardware-software integration. Passionate about developing reliable, efficient, and robust embedded solutions for automotive and real-time applications.

PROFESSIONAL EXPERIENCE

SERMA International Embedded Software Engineer - Developed a C++ and Qt application to automate electronic equipment testing. - Develop Python drivers to control and automate laboratory equipment used in the automated test bench. - Integrated 15+ ATP (Acceptance Test Procedure) tests into the application and validated their execution. - Development of a Python script to automatically generate ATR (Acceptance Test Results). Keywords: C++, Python, Qt	04/2026 – Present Ariana, Tunisia
Automotive Test and Validation Engineer - Configured and tested Hardware-in-the-Loop (HIL) A-IVI automotive systems. - Executed functional, non-functional, regression, non-regression, performance and stress testing. - Validated infotainment functionalities including Apple CarPlay , Android Auto and ADAS system. - Applied black-box and exploratory testing techniques to identify and isolate defects. - Designed and executed test cases and test scenarios using CodeBeamer and Silk Central. - Analyzed audio quality and system behavior using R&S® UPP. - Tracked and triaged anomalies using Jira, including SQL-based dashboards and ticket monitoring. - Created and maintained 10–15 bug tickets and user stories each month across development and QA cycles. - Participated in client meetings to review requirements, anomalies, and software-release stability. - Conducted audio analysis at the end of each sprint and reported audio-quality status for software releases and new features. Keywords: Vector CANalyzer, Silk Central, CodeBeamer, Jira, R&S® UPP, DDT2000, HIL, Agile, Automotive ECU	08/2025 – 03/2026 Ariana, Tunisia
End-of-Studies Internship - Developed a Python-based AI system to automate and improve Jira ticket management within an automotive test and validation environment. - Used transformer models such as BERT, SBERT, RoBERTa, and BART for ticket classification, duplicate detection, specification verification, and test-log summarization. - Automated the generation, validation, prioritization, and quality assessment of Jira tickets from test results and technical logs. - Integrated the AI functionalities into a user-friendly application using Python and PySide6/QML to support the test and validation workflow. Keywords: Python, NLP, DL, NLTK, OCR, JIRA, QML	02/2025 – 07/2025 Ariana, Tunisia
ACTIA Engineering Services Engineering Internship - Developed an embedded air-quality monitoring system using Rust on STM32 and ESP32 microcontrollers. - Implemented real-time sensor data acquisition and processing for air-quality monitoring. - Integrated Bluetooth, UART, and MQTT communication protocols for reliable data transmission. - Developed real-time data visualization to monitor and display sensor measurements remotely. Keywords: STM32, ESP32, AQS, Linux, Rust, UART, MQTT	07/2024 – 09/2024 Ariana, Tunisia

EDUCATION

Faculty of Sciences of Tunis, University of Tunis El Manar National Engineering Diploma in Electronics.	2022 – 2025 Manar, Tunisia
Preparatory institute for engineering studies of Kairouan Pre-Engineering Diploma, Mathematics and Physics	2020 – 2022 Kairouan, Tunisia

SKILLS

Design and Simulation software:

Proteus (ISIS), MATLAB/SIMULINK, Vector CANalyzer, DDT2000, Qt Creator

Programming Languages:

Rust, C, C++, C#, Python

Soft Skills:

Project Planning, Problem-Solving, Time Management, Teamwork

Operating Systems:

Linux (Ubuntu)

Communication Protocol:

CAN, I2C, UART, SPI, MQTT

Electronic Cards:

STM32, Raspberry Pi, ESP32, Arduino, NI6363, NI6501

Testing Techniques:

Black-Box, White-Box, Exploratory Testing

CERTIFICATES

EF SET : English certificate

C++ Programming: Classes and Data

Introduction and Programming with IoT Boards

LANGUAGES

English B2	French B2	Arabic Langue maternelle
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